

Problem 2.5

\$100 is deposited at time $t = 0$ into an account whose accumulation function is

$$a(t) = 1 + 0.03\sqrt{t}.$$

1. Find the amount of interest generated at time 4, i.e., between $t = 0$ and $t = 4$.
2. Find the amount of interest generated between time 1 and time 4.

Solution

Since the initial deposit is \$100, the accumulated amount at time t is

$$A(t) = 100a(t).$$

(a) Interest generated between $t = 0$ and $t = 4$:

$$I_{[0,4]} = A(4) - A(0) = 100(a(4) - a(0)).$$

Now,

$$a(4) = 1 + 0.03\sqrt{4} = 1 + 0.06 = 1.06,$$

and

$$a(0) = 1 + 0.03\sqrt{0} = 1.$$

Therefore,

$$I_{[0,4]} = 100(1.06 - 1) = 100(0.06) = 6.$$

Hence,

$$\boxed{I_{[0,4]} = \$6}.$$

(b) Interest generated between $t = 1$ and $t = 4$:

$$I_{[1,4]} = A(4) - A(1) = 100(a(4) - a(1)).$$

We already know that $a(4) = 1.06$, and

$$a(1) = 1 + 0.03\sqrt{1} = 1.03.$$

Thus,

$$I_{[1,4]} = 100(1.06 - 1.03) = 100(0.03) = 3.$$

Therefore,

$$\boxed{I_{[1,4]} = \$3}.$$